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[22EC522]

# FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

## RFID and FLEXIBLE SENSORS

**Time: 3 Hours** **Maximum Marks: 100**

**Instructions:** i. Missing data may be suitably assumed.

**ANSWER ALL QUESTIONS**

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| | | **PART-A** | | | **5 x2=10** |
| **1.** | **a.** | In case of FasTag RFID system, specify the frequency, range (approx.), power source (type), data rate ( low/medium/high). | L4 | CO1 | **02** |
| | **b.** | Discuss the payback period in supply chain management using RFID. | L3 | CO2 | **02** |
| | **c.** | What are the ethical issues related to RFID Technology. | L3 | CO4 | **02** |
| | **d.** | What is a non invasive sensor, explain with an example. | L3 | CO5 | **02** |
| | **e.** | Define gain and directivity of an antenna. | L3 | CO6 | **02** |
| | | **PART-B** | | | **6 X 5=30** |
| **2.** | **a.** | Discuss the functionalities of interrogator and discuss the following feature of reader<br>**i.** Anticollision<br>**ii.** Authentication<br>**iii.** Data security | L3 | CO1 | **05** |
| | | **OR** | | | |
| | **b.** | Discuss read only (RO), Read write (RW), WORM, and smart tags with applications and examples. | L3 | CO1 | **05** |
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| **3.** | **a.** | Discuss the improvement in supply chain metrics by adopting RFID technology. | L3 | CO2 | **05** |
| | | **OR** | | | |
| | **b.** | What is middleware of RFID system? Discuss in detail the 4 main functions of the Middleware. | L3 | CO3 | **05** |
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| **4.** | **a.** | What is the end user, knowledge, experience, confusion and scepticism with respect to RFID technology. | L3 | CO4 | **05** |
| | | **OR** | | | |
| | **b.** | What are privacy concerns of RFID Technology and discuss the RFID industry responds to privacy concerns. | L3 | Co4 | **05** |
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| **5.** | **a.** | Name the sensors used for measuring the following physical variable.<br>**i.** Motion/movement<br>**ii.** Angular velocity<br>**iii.** Liquid pressure<br>**iv.** Ground velocity.<br>**v.** light | L3 | CO5 | **05** |
| | | **OR** | | | |
| | **b.** | Discuss different methods of calibration of sensor | L3 | CO5 | **05** |

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| **6.** | **a.** | Discuss the dipole antenna structure , features and radiation pattern with figures. | L3 | CO6 | **05** |
| | | **OR** | | | |
| | **b.** | Discuss the design of high directivity RFID antenna with relevant figures | L4 | CO6 | **05** |
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| **7.** | **a.** | Discuss the features and modes of operation of micro controller module for RFID enabled sensor system. | L3 | CO6 | **05** |
| | | **OR** | | | |
| | **b.** | Illustrate dipole antenna with stubs and examine performance through radiation pattern | L3 | CO6 | **05** |
| | | **PART-C** | | **6 X 10=60** | |
| **8.** | **a.** | Compare and contrast the barcode system and RFID technology with justification. | L5 | CO1 | **10** |
| | | **OR** | | | |
| | **a.** | Discuss the features of low frequency (LF/HF) operation of RFID with examples. | L3 | CO1 | **05** |
| | **b.** | "The controller is termed the brain of RFID system" Substantiate the statement. | L4 | CO1 | **05** |
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| **9.** | **a.** | Discuss the operations of the following blocks of the middleware<br>**i.** Core processing<br>**ii.** Reader interface<br>**iii.** Enterprise application adapter | L3 | CO3 | **08** |
| | **b.** | What is middleware of RFID system? Where it is located in the RFID system | L3 | CO3 | **02** |
| | | **OR** | | | |
| | **a.** | Discuss the middleware architecture as apart of the RFID system with a diagram | L3 | CO3 | **06** |
| | **b.** | List at least two applications of RFID technology in the following commercial sector.<br>**i.** Transportation and distribution<br>**ii.** Retail and consumer packaging | L4 | CO4 | **04** |
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| **10.** | **a.** | Illustrate and discuss EPC network with a schematic diagram. | L3 | CO4 | **06** |
| | **b.** | What is a kill switch; elaborate its requirement y with an example. | L4 | CO4 | **04** |
| | | **OR** | | | |
| | **a.** | Substantiate the statement " The RFID is immature technology for widespread adoption" | L4 | CO4 | **05** |
| | **b.** | What is the need for RFID regulations and discuss the Govt. regulations for RFID | L3 | CO4 | **05** |

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| **11.** | **a.** | Discuss the following groups of sensors with example<br>**i.** Contact and non contact sensors<br>**ii.** Absolute and relative sensors | L3 | CO5 | **04** |
| | **b.** | Compare<br>**i.** analog and digital sensors<br>**ii.** active and passive sensors | L5 | CO5 | **06** |
| | | **OR** | | | |
| | **a.** | Discuss the following parameters with respect to sensor with examples.<br>**i.** Sensitivity<br>**ii.** Resolution<br>**iii.** Accuracy<br>**iv.** Linearity | L3 | CO5 | **06** |
| | **b.** | Identify and explain the measurable physical variable of the following sensors<br>**i.** Proximity sensor<br>**ii.** NaVIC/GPS | L3 | Co5 | **04** |
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| **12.** | **a.** | Discuss the performance of Bowtie T matched RFID antenna with structure, reflection coefficient and radiation pattern figures. | L3 | CO6 | **08** |
| | **b.** | Define reflection coefficient of an antenna. | L3 | CO6 | **02** |
| | | **OR** | | | |
| | **a.** | Discuss wireless transmitter of RFID enabled sensor with system level diagram. | L3 | CO6 | **05** |
| | **b.** | Illustrate dipole antenna with stubs and examine performance through radiation pattern. | L3 | CO6 | **05** |
| | | | | | |
| **13.** | **a.** | Discuss the government initiatives for extending the. applications of RFID technology | L3 | Co4 | **05** |
| | **b.** | Discuss the privacy risks of RFID technology and measures to mitigate the same. | L3 | CO4 | **05** |
| | | **OR** | | | |
| | **a.** | What are the challenges in the data management of RFID technology? | L3 | CO4 | **06** |
| | **b.** | Discuss the wireless sensor network architecture with respect to RFID with a figure. | L3 | CO5 | **04** |